

Integration of repellency effect of neem-based insecticide and pheromone Bio-Trap[®] with *Beauveria bassiana* (Hypocreales: Cordycipitaceae) to control the red palm weevil, *Rhynchophorus ferrugineus* (Olivier) (Coleoptera: Curculionidae)

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To control adults of the red palm weevil (RPW), *Rhynchophorus ferrugineus* (Olivier) (Coleoptera: Curculionidae), a pheromone Bio-Trap[®] containing the commercial formulation of *Beauveria bassiana* (Broadband™) was integrated with neem-based insecticide, (azadirachtin). *Beauveria bassiana* at 40 ml suspension of 1×10^9 spores/ml caused 100% mortality of RPW. The LC₅₀ of neem-based insecticide, (azadirachtin) to RPW by direct spraying was 1810.3 ppm, whereas that by feeding was 2117.80 ppm. The lowest concentration of neem that was 100 % repellent to RPW was 800 ppm. The repellency effect of neem was integrated with the pheromone Bio-Trap in three net houses (5 × 5 m), each planted with five offshoots of palm trees (height, 1–1.5 m). In two of the net houses, wounds were made on offshoots and sprayed weekly with 800 ppm neem, and a pheromone Bio-Trap was introduced. The third net house was the control, in which wounds were made on the offshoots without the spraying of neem and a pheromone non-Bio-Trap (without fungus) was introduced. Thirty adults were introduced, and after 30 days in the net houses with the pheromone Bio-Trap, all RPW died because of contamination by fungi. The repellency of neem prevented the insects from laying eggs in the wounded offshoots, and no larvae were found. However, in the control net house, only 60 % of the adults were trapped and killed in the pheromone non-Bio-Trap, and two of the five offshoots were infested with 26–34 larvae.

Key words: botanical insecticide, azadirachtin, IPM, biological agents, invasive insect.

INTRODUCTION

Date palm is the main crop in countries of the Arabian Gulf and the production of dates is an important part of the food security, particularly for rural communities. Culturally, dates are one of the foods consumed daily in these arid regions. The red palm weevil (RPW), *Rhynchophorus ferrugineus* (Olivier) (Coleoptera: Curculionidae), is an invasive species that attacks date palms, which was discovered in the Arabian Gulf countries in mid-1985 and in Saudi Arabia in early 1987 (Al-Abdulmohsin 1987). Because the pest management tools were insufficient and phytosanitary measures were weakly implemented, RPW spread rapidly in Saudi Arabia, primarily through infested plant materials transported from one area to another (Faleiro *et al.* 2019). The RPW has become the most destructive pest affecting palm trees, causing serious damage that ultimately leads to the complete removal of infested palm

trees. As a result, the numbers of palm trees are decreasing in the countries of the Arabian Gulf (Faleiro 2006). This pest is expanding its geographical range, becoming a worldwide invasive pest species in the last four decades. The RPW has been recorded in the Middle East, Europe, North Africa, and the Caribbean Islands (Faleiro 2006; Giblin-Davis *et al.* 2013; Faleiro *et al.* 2019).

Chemical pesticides are the most common tool used to control pests and weeds in both undeveloped and developed countries (Whitten & Oakeshott 1991). Pesticides are widely used because they are toxic to a wide range of pests, and are cheap and easily applied using spraying equipment. Globally, \$25–30 billion is spent annually on pesticides (Pretty & Hine 2005). In Saudi Arabia, approximately 3700 t of pesticides are used annually (pers. comm., Ministry of Agriculture, 2018). Although chemical pesticides are effective in



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protecting crops, there is a concern about negative effects on the environment and human health. Because of the intensive use of chemical pesticides, RPW has recently developed resistance to the commonly used insecticides such as organophosphorus and synthetic pyrethroid insecticides, impairing the efficacy of RPW control in the field (Al-Ayedh *et al.* 2016; Wakil *et al.* 2018). Therefore, alternative techniques to protect plants, including botanical insecticides and biological methods (Pretty & Hine 2005) are necessary.

The weevils of *Rhynchophorus* genus are attacked by more than 50 different natural enemies, including viruses, bacteria, yeast, fungi, nematodes, mites, insects, and vertebrates (Mazza *et al.* 2014). Both fungi and nematodes are promising biological control agents against RPW (Dembilio *et al.* 2010; Manachini *et al.* 2013; Hussain *et al.* 2017). The insect pathogenic fungi, *Beauveria bassiana* and *Metarhizium anisopliae* can infect and kill a variety of arthropod species. Both fungi are used as insect pathogens in the biocontrol of agricultural pests. In particular, *M. anisopliae* is used to control grasshoppers and locusts in different parts of the world. Contact with or intake of these two entomopathogenic fungi reduces the populations of target insects (Bateman *et al.* 1996; Clarkson & Charnley 1996; Wraight *et al.* 2010; Hussain *et al.* 2017). In particular, the entomopathogenic fungus, *B. bassiana* has shown great promise in the control of RPW (Güerri-Agulló *et al.* 2011). Laboratory and semi-field cage studies show the possibility of infecting and killing RPW adults with *B. bassiana* by using pheromone traps (Hajjar *et al.* 2015). However, hot and dry conditions are very unfavourable towards the implementation of entomopathogenic fungi, which require warm and moist conditions. Faleiro *et al.* (2019), in an overview of the gaps, challenges, and prospects of RPW management, found that the known entomopathogenic agents (fungi and nematodes) are not adequately exploited in the control of the RPW. In Saudi Arabia and the Arabian Gulf countries, the use and implementation of these agents in date palm fields have not been promising, which is undoubtedly due to the extremely hot and dry conditions that prevail in the arid regions of the world in which date palms are cultivated. In addition, larval stages in the life cycle of the RPW are hidden in the trunks of date palms and the use of biological agents in the field is challenging (Al-Dosary *et al.* 2016). Therefore,

insecticides remain the main tool used to control this pest, in addition to some cultural protective measures (Faleiro 2006). In developing an integrated pest management (IPM) programme to control the RPW, an important alternative to the commonly used insecticides is the use of a botanical insecticide that is less harmful to the environment and humans. The botanical insecticide, neem-based azadirachtin is an alternative, environmentally friendly insecticide that is used to control many harmful pests. The active substance azadirachtin is extracted from neem seeds, producing an organic insecticide that has strong effects on pest behaviour, with antifeedant and repellent effects affecting pest growth (Schmutterer 1990; Ruckin 1992; Lowery & Isman 1994; Tunca *et al.* 2014). Neem extracts affect more than 200 species of insects, including white flies, thrips, beetles, and moths (Miller & Uetz 1998). Various malformations were recorded on adult weevils treated with azadirachtin. A negative correlation was observed between the concentration of azadirachtin and the rate of adult emergence of RPW (Bream *et al.* 2001; Abahussain 2008). Neem is primarily injected into infested palms, and this method of application is effective in killing various stages of the RPW in the trunks of palms, thereby reducing the infestation (Ghoneim *et al.* 1998; Azam *et al.* 2001; Abahussain 2008; Sujatha *et al.* 2009). Furthermore, Al-Shawaf *et al.* (2013) reported that azadirachtin kills 96.7 % of RPW eggs in treated sugarcane pieces. In addition, Merghem & Mohammad (2017) reported the potential repellent effect of neem on other insects and especially on the RPW. The study objectives were to evaluate the integration of a pheromone Bio-Trap[®] with *B. bassiana* with the repellent effect of neem on trunks of date palms to kill RPW adults with entomopathogenic fungus and prevent oviposition in trunk wounds treated with neem.

MATERIAL AND METHODS

Test insects

The adults and larvae of RPW were obtained from the laboratory of the Date Palm Research Centre of Excellence (DPRCE), King Faisal University, Hofuf, Saudi Arabia. Pineapple was used as the food medium because it is attractive to females for oviposition. All larvae were reared in the laboratory (25–30 °C) on pineapple in cages (45 × 38 × 38 cm) made of transparent polyethylene

with wire mesh on the top, whereas the adults were maintained in cages on fresh palm fronds and dates.

Test materials

Becker Underwood, Inc. (West Sussex, U.K.) supplied the liquid formulation of spores suspension at 1×10^9 spores/ml of *B. bassiana* (Broadband™), Astrachem, Saudi Arabia, supplied Azdar 10 EC, the neem-based botanical insecticide, azadirachtin, ChemTica International, S.A., Costa Rica, supplied the ferrugineol pheromone lure

Pheromone Bio-Trap® of the RPW

The RPW trap was designed as described by Hajjar *et al.* (2015) to attract and contaminate RPWs with the conidia of *B. bassiana* from the commercial product (Broadband™). The transfer of the conidia from the infected (directly exposed) adults to uninfected adults was expected during mating (Hajjar *et al.* 2015). The trap was a small bucket (height 25 cm; diameter of circular bottom 15 cm) with four symmetric holes (4×4 cm) to allow entry for beetles. Food bait, pieces of palm fronds, and date fruits were placed in the bottom of the bucket. The ferrugineol pheromone lure (ChemTica International, S.A., Costa Rica) was hung internally from the bucket lid. A layer of sackcloth that covered the internal surface of the bucket was soaked with 40 ml of Broadband™ suspension at 1×10^9 spores/ml. The moist sackcloth in the bucket helped the fungus to overcome the extremely hot and dry conditions that exist in Saudi Arabia.

Preliminary test to prove the pathogenicity of fungi

The dead red palm weevils collected from the net house with Bio-Trap were put into sterile Petri dishes with high humidity and incubated at 25 ± 1 °C for 72 h, in complete darkness to retrieve fungal spores and confirm fungal infection.

Neem bioassays

The neem botanical insecticide was tested in a range of dose-response bioassays against adult RPWs. Six serial dilutions of neem were tested, ranging from 500 to 5000 ppm; controls were treated with distilled water only. Three replicates of 10 adults (five males + five females) were treated topically with each dilution by using a spray chamber, which was fitted with an air-atomising nozzle connected to a regulator valve providing a constant airflow of 10 l/min. The target

surface in the spray chamber was a rotating turntable (30 rpm).

In addition, a bioassay test was conducted on the oral administration of neem by using treated pieces of palm frond as food. Three replicates were used for each dilution, and each replicate was 10 adults (five males + five females) placed with treated food. Bioassays were conducted at ambient laboratory temperature (25 ± 2 °C), and mortality was recorded after 24 h for the topical application (direct spray) and daily for 14 days for the oral administration. Adults with no signs of movement were recorded as dead. Probit statistical analysis was used to determine the LC_{50} and LC_{90} values of treated insects (PoloPlus software program).

Repellency effect of neem on the RPW

This test was carried out to determine the concentration of neem that could overcome the effect of other attractive factors to RPW. The attractive factor to the weevil (pheromone) was the same in all treatments, but the concentration of the neem-based insecticide, was variable. Filter papers were placed in glass Petri dishes (20 cm); a piece of treated frond and date fruit was added. In the control dishes, food was treated with distilled water. The pheromone was placed in the dishes of both the treatments and controls. In a glass box ($30 \times 30 \times 20$ cm), the Petri dishes with treated frond and date were placed at one end and the control dishes at the opposite end. To determine the lowest concentration repellent to the RPW, eight concentrations of neem below the value of the LC_{50} were prepared, starting from 100 ppm and increasing in increments of 100 ppm to 800 ppm. Four boxes were used as replicates for each concentration, and 20 numbered labelled adults were placed inside each box in the middle between treated and control Petri dishes. The tests were conducted at ambient laboratory temperature (25 ± 2 °C). The behaviour of the weevils was monitored daily for at least 6 h by using a digital video camera (Alpha a7 II Mirrorless Digital Camera with FE 28–70 mm f/3.5–5.6 OSS lens, Sony, Japan) and the number of adults that avoided movement across to the treated food under different concentrations of the neem was recorded for 6 h.

Persistence of the repellent effect of neem

This trial was conducted to determine the period for which a concentration of neem could repel the

RPW. The lowest repellent concentration that led to 100 % repellency of the RPW was used. Twenty unused marked weevils were placed in each replicate (glass box) on a daily basis for eight days, and the same treatment and control Petri dishes were used from the first day of the test. This bioassay was also conducted at ambient laboratory temperature (25 ± 2 °C). The duration of the repellent action of the tested concentration, and the repellent effect over time were determined by monitoring weevil behaviour. The number of adults avoiding movement across to the treated food under different concentrations of the neem was recorded for at least 6 h by using a digital video camera (Alpha a7 II Mirrorless Digital Camera with FE 28–70 mm f/3.5–5.6 OSS lens, Sony, Japan) and for a duration of 8-day post treatment

Integration of the repellency effect of neem with the pheromone Bio-Trap of B. bassiana to control the RPW

This trial was conducted in the field and integrated the efficacy of the Bio-Trap with *B. bassiana* (Broadband™) in attracting, infecting, and killing the RPW as described by Hajjar *et al.* (2015). Four net houses (5 × 5 m) were used, and five palm trees (height, 1–1.5 m) were planted in each house. Cuts were made in the trunks to artificially wound the palms. In two of the houses, then the whole five trees of each net house were sprayed weekly with 10 l of neem solution as a repellent at the concentration of 800 ppm, and the pheromone Bio-Trap was placed in the middle of each house. The other two net houses were considered as the control. The common bucket pheromone trap without the fungus, that is recommended by Agriculture authority in KSA, was placed in the middle of the control net house, which contained in the bottom of the bucket an attractive food bait of palm frond pieces and date fruits that was mixed with the insecticide cypermethrin. The palms in the control net houses were not sprayed. Thus, in both treatment and control, the ferrugineol pheromone lure was hung internally from the bucket lid. In each net house, 30 adult beetles (15 males + 15 females) were introduced. The effectiveness of the treatment integrating the repellent concentration of neem and the pheromone Bio-Trap of *B. bassiana* was evaluated under the prevailing environmental conditions of 40–48 °C in Alhasa, Saudi Arabia. Survival was recorded daily for 30 days after introduction, in

addition to tracking the presence of RPW larvae in the wounds on trunks by thoroughly searching for larvae and their damage symptoms.

Statistical analyses

The lethal concentration that caused 50 % mortality (LC₅₀) was determined by probit analysis using the PoloPlus software program (Le Ora Software, Berkley, California) (Robertson *et al.* 2003).

Daily and cumulative mortality caused by the combination of the neem treatment and the pheromone Bio-Trap were subjected to quadratic polynomial and regression models ($y = ax^2 + bx$), fitted between the individual dose of bio-insecticide response data and the corresponding individual interval time (days) (Carlberg 2018). The number of RPW larvae found in palm trees were subjected to analysis of variance (ANOVA), and the means were compared by the least significant differences (LSD) at $P \geq 0.05$ using the SAS 9.3 statistical software package (2007).

RESULTS

In the preliminary test, the commercial fungus (Broadband™) infected RPW by contact. The morphological characteristics of mycelia and spores retrieved from dead weevils confirmed infection by the fungal species *B. bassiana*.

Neem bioassays

According to the probit analysis, the LC₅₀ of topical application (direct spray) was 1810.3 ppm and that of oral administration was 2117.8 ppm and the two LC₅₀ values were not significantly different (Table 1). The LC₉₅ values were 6855.6 ppm with topical application and 6947.8 ppm with oral administration, which were also not significantly different at the confidence limit (CL) of 95 %. However, the value of the slope of the LCP line was greater with oral administration than with topical application, and the value of the *R* factor was higher with topical application (0.982) than with oral administration (0.926), but there was no significant difference ($P < 0.05$) between the slope values of topical and oral application.

Repellent effect of neem

The repellent effect of neem was determined based on concentration below the LC₅₀ values. The number of weevils repelled increased as the concentration of the insecticide increased; the

Table 1. LC₅₀ and LC₉₅ values (ppm) of the neem insecticide for *Rhynchophorus ferrugineus*.

Treatment	Slope ± S.E.	LC ₅₀ (ppm)	95% CL ***	LC ₉₅ (ppm)	95% CL ***	Chi-square	R
Topical application*	2.84 ± 0.52	1810.3	1346.5–2323.3	6855.6	4698.5–14149.5	0.142	0.98
Oral administration**	3.19 ± 0.59	2117.8	1629.6–2664.8	6947.8	4907.8–13596.8	7.840	0.93

*Topical application was done in a spray chamber, and mortality was recorded after 24 h.
**Oral administration was through contact with and oral ingestion of treated pieces of palm fronds, and accumulative mortality was recorded over a 14 d period.
***Upper and lower limit at CL 95 %.

repellency of neem began at 400 ppm and continued up to 800 ppm (Fig. 1). The repellency effect at 800 ppm was 100 %, which lasted for 2 days of treatment. The efficiency of repellency then decreased with time, and 80 % of the weevils were repelled from the treated surface at 3 days, 60 % at 4 days, 40 % at 5 days, 25 % at 6 days, and 5 % at 7 days.

Integration of the repellency effect of neem with the pheromone Bio-Trap of *B. bassiana* to control RPW

This trial studied the effectiveness of the pheromone Bio-Trap in the field to kill weevils by

contaminating them with spores of the fungus *B. bassiana*, in addition to the possibility of transferring the spores to other weevils during mating as indicated by Hajjar *et al.* (2015). Additionally, the effectiveness of neem as a repellent in preventing weevils from laying eggs on palms was examined by creating artificial wounds on the palm trunks. However, at the lowest concentration of neem that was 100 % repellent to adult RPWs (800 ppm), the 100 % repellency effect persisted for 2 days. Therefore, in the field trial, neem at 800 ppm was sprayed weekly to obtain a longer effective period in repelling weevils from treated palms.

The results from the two treated net houses with

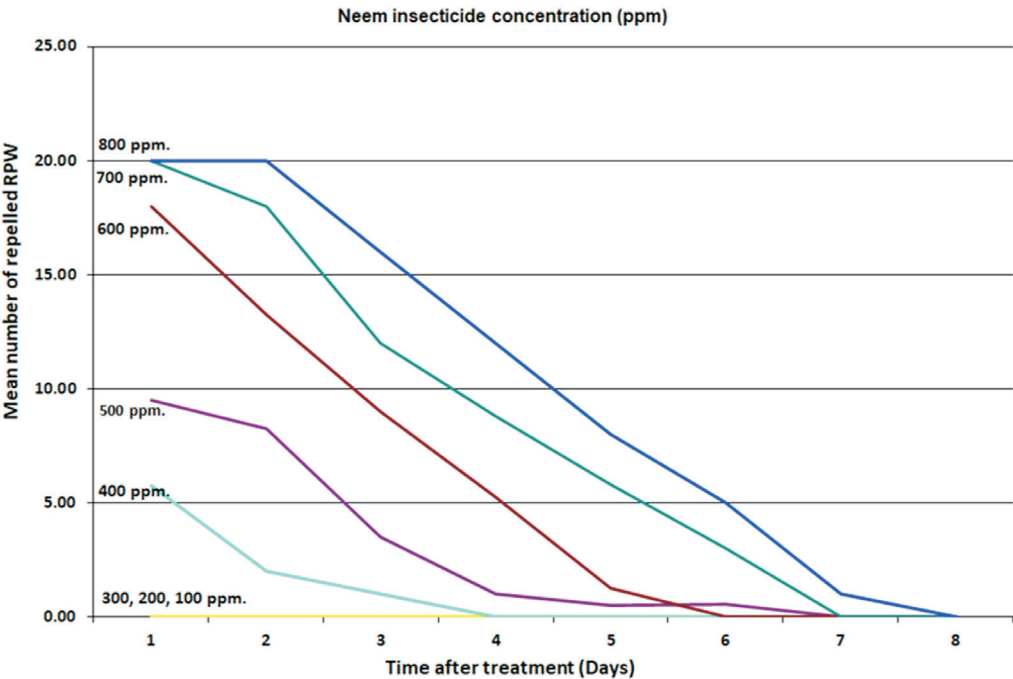


Fig. 1. Mean number of *Rhynchophorus ferrugineus* individuals repelled at different concentrations (ppm) of the neem insecticide over an 8-day period. The numbers of repelled adults were recorded daily from monitoring the insects in the boxes for 6 h daily by using a digital video camera; all values represented as mean of 20 adults insect; no. of replicates = 4.

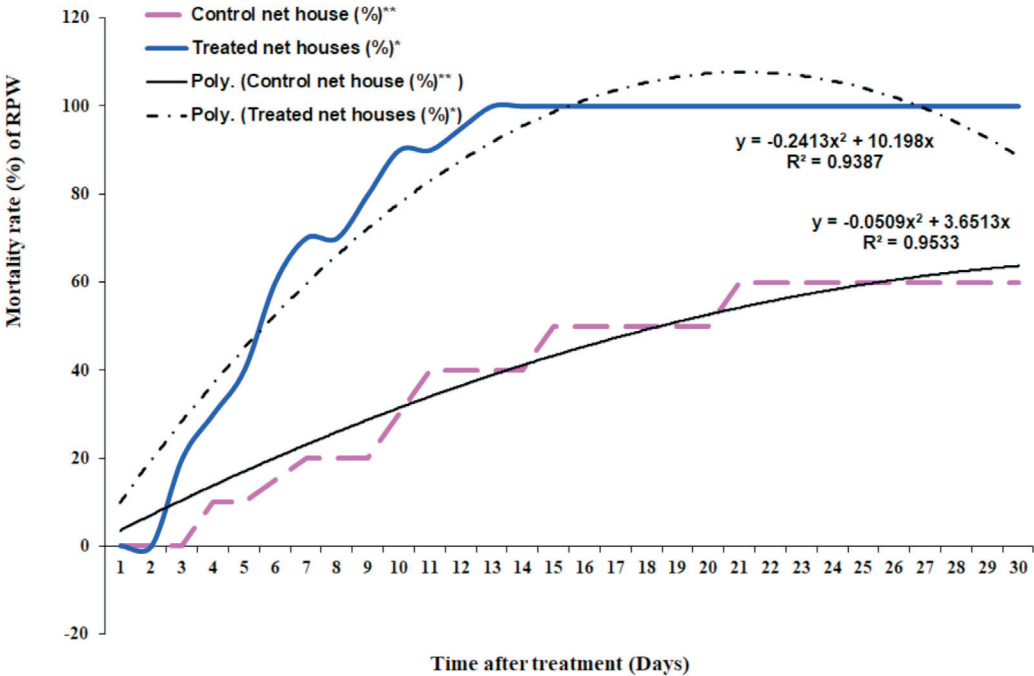


Fig. 2. Mortality rate (%) of *Rhynchophorus ferrugineus* infected with *Beauveria bassiana* in the treated net houses with a Bio-Trap and in the control net house with a pheromone trap over a 30-day period. *Treatment; % of mean death, $n = 30$, no. of replicates in treated net house = 2. **Control % of death of $n = 30$.

the introduction of the pheromone Bio-Trap of *B. bassiana* are shown in Fig. 2. The accumulative death rate was 100 % after 13 days, and the cause of death was infection by the spores of *B. bassiana* from the pheromone Bio-Trap ($r^2 = 0.9387$). To confirm death by the fungus and retrieve the mycelia, the dead adults were transferred to sterile Petri dishes and incubated at $25 \pm 1^\circ\text{C}$ and $75\% \pm 5\%$ humidity. By contrast, in the control net house, the accumulative death rate reached only 60 % after 30 days ($r^2 = 0.9533$), and all the dead weevils were attracted to the pheromone in the common trap used by Agriculture authority in KSA, and killed by insecticide mixed with food, in the trap.

In addition to determining the effect of using the pheromone Bio-Trap, the repellent effect of neem to protect palms was studied by measuring the ability of weevils to lay eggs in wounded and treated trunks after 30 days. The high repellency of neem prevented females from laying eggs on the treated trunks, and compared with the palms in the control net house, the trunks of treated palms were 100 % free of infestation by RPW larvae. However, in each of the control net houses,

four of the five palms had RPW larvae in the trunk wounds (Table 2). The mean number of larvae found in the control was significantly higher than the treatment with neem at the repellent concen-

Table 2. Number of *Rhynchophorus ferrugineus* of larvae after 30 days in wounded palms grown in net houses. Palms were either treated with neem at a repellent concentration of 800 ppm or not treated (control)

Palm tree	No. of larvae			
	Control net house		Treatment net house	
	No. 1	No. 2	No. 1	No. 2
1	15	0	0	0
2	12	15	0	0
3	9	26	0	0
4	24	17	0	0
5	0	7	0	0
Mean	12.5 ^a ± 2.8		0.00 ^b ± 0.00	
L.S.D (0.05)	* 5.07			

Mean represented for 10 trees $\pm$ SE.
*Significant values at $F = 20.018$, d.f. = 9, $P = 0.0003$.
LSD: Lowest significant differences at $P \geq 0.05$.
Means in the same row without the same alphabetical letter are significantly different at $P = 0.05$.

tration of 800 ppm with the pheromone Bio-Trap ($F = 20.018$, d.f. = 9, $P = 0.0003$).

DISCUSSION

Neem insecticide is used to control the RPW either as a direct spray or as an injection in the trunk of palms (Bream *et al.* 2001; Abahussain 2008; Sujatha *et al.* 2009; Al-Shawaf *et al.* 2013; Porcelli *et al.* 2013; Merghem & Mohamed 2017). The repellent effect of neem to the RPW is rarely studied. This study may be the first to investigate the repellency effect of sub-lethal doses of neem to the RPW, where the concentration of 800 ppm caused 100 % repellency to adults for 2 days after treatment and then the effect declined to 80 % after 3 days and to 40 % after 5 days. The repellency effect of neem extract to RPW was also reported by Merghem & Mohamed (2017), with both remedial and protective effects, against RPW in field trials in Egypt. The latter study reported a considerable reduction of infestation as results of average repellency from 53.15 to 60.8 % for all applied treatments of neem extracts. In addition, the antifeedant and repellency effects of neem have been studied in many insects. According to Isman (2006), neem remains the most effective anti-feedant agent against desert locust. A sublethal dose of azadirachtin have also been reported to affect the development of insects such as *Earias vittella* (Fab.) (Lepidoptera: Nolidae) (Gulzar *et al.* 2019), and the emergence ratio during development and the longevity of beneficial insects such as *Chelonus oculator* (F.) (Hymenoptera: Braconidae) (Tunca *et al.* 2014) and *Habrobracon hebetor* Say (Hymenoptera: Braconidae) (Abedi *et al.* 2014).

In this study, neem repellency together with exposure to *B. bassiana* in pheromone traps achieved good control in a field control programme of RPW. Whereas in the 30 day field test, neem applied onto palm tree trunks at the repellent concentration of 800 ppm, in combination with the

pheromone Bio-Trap of *B. bassiana*, was highly effective in preventing females from laying eggs in the wounded trunks of trees, when compared with untreated trees in the control. The high efficacy of the pheromone Bio-Trap of *B. bassiana* in the field trial is consistent with the finding of Güerri-Agulló *et al.* (2011), who reported great promise with *B. bassiana* for the control of RPW. Similarly, laboratory and semi-field cage studies indicated that the pheromone Bio-Trap of *B. bassiana* is effective in infecting and killing RPW (Hajjar *et al.* 2015). In addition, the field repellency effect of neem was also compatible with the repellency effect of neem to RPW reported by Merghem & Mohamed (2017). The results of this study showed that the application of *B. bassiana* could be adequately exploited to control RPW by keeping the fungus moist through lining the bucket of the pheromone Bio-Trap with sackcloth. The latter helped to overcome the extremely hot and dry conditions that exist in the arid regions of the world, where date palms are cultivated.

CONCLUSIONS

In this study, the integration of the pheromone Bio-Trap of *B. bassiana* with the repellent effect of a sub-lethal concentration of neem sprayed on palms under field conditions was effective in the control of the RPW, killing all weevils. Neem effectively repelled the RPW and prevented the females from laying eggs in wounded and treated palms.

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